
STM32
CubeMX

1. Description

1.1. Project

Project Name	FRW4_0_RTOS
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	02/10/2026

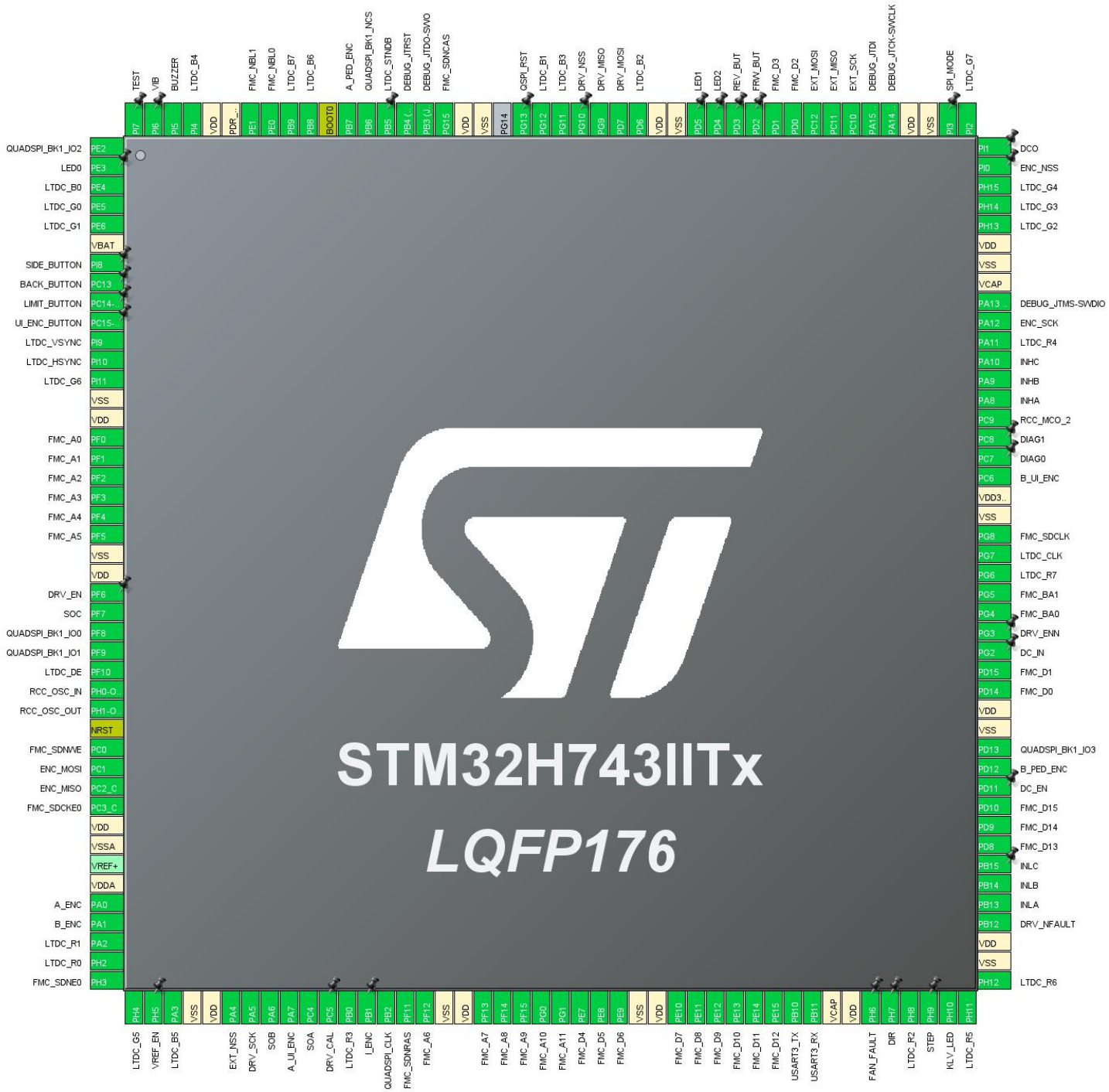
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H743/753
MCU name	STM32H743IITx
MCU Package	LQFP176
MCU Pin number	176

1.3. Core(s) information

Core(s)	ARM Cortex-M7

2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	QUADSPI_BK1_IO2	
2	PE3 *	I/O	GPIO_Output	LED0
3	PE4	I/O	LTDC_B0	
4	PE5	I/O	LTDC_G0	
5	PE6	I/O	LTDC_G1	
6	VBAT	Power		
7	PI8	I/O	GPIO_EXTI8	SIDE_BUTTON
8	PC13	I/O	GPIO_EXTI13	BACK_BUTTON
9	PC14-OSC32_IN (OSC32_IN)	I/O	GPIO_EXTI14	LIMIT_BUTTON
10	PC15-OSC32_OUT (OSC32_OUT)	I/O	GPIO_EXTI15	UI_ENC_BUTTON
11	PI9	I/O	LTDC_VSYNC	
12	PI10	I/O	LTDC_HSYNC	
13	PI11	I/O	LTDC_G6	
14	VSS	Power		
15	VDD	Power		
16	PF0	I/O	FMC_A0	
17	PF1	I/O	FMC_A1	
18	PF2	I/O	FMC_A2	
19	PF3	I/O	FMC_A3	
20	PF4	I/O	FMC_A4	
21	PF5	I/O	FMC_A5	
22	VSS	Power		
23	VDD	Power		
24	PF6 *	I/O	GPIO_Output	DRV_EN
25	PF7	I/O	ADC3_INP3	SOC
26	PF8	I/O	QUADSPI_BK1_IO0	
27	PF9	I/O	QUADSPI_BK1_IO1	
28	PF10	I/O	LTDC_DE	
29	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
30	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
31	NRST	Reset		
32	PC0	I/O	FMC_SDNWE	
33	PC1	I/O	SPI2_MOSI	ENC_MOSI
34	PC2_C	I/O	SPI2_MISO	ENC_MISO

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
35	PC3_C	I/O	FMC_SDCKE0	
36	VDD	Power		
37	VSSA	Power		
39	VDDA	Power		
40	PA0	I/O	TIM2_CH1	A_ENC
41	PA1	I/O	TIM2_CH2	B_ENC
42	PA2	I/O	LTDC_R1	
43	PH2	I/O	LTDC_R0	
44	PH3	I/O	FMC_SDNE0	
45	PH4	I/O	LTDC_G5	
46	PH5 *	I/O	GPIO_Output	VREF_EN
47	PA3	I/O	LTDC_B5	
48	VSS	Power		
49	VDD	Power		
50	PA4	I/O	SPI3_NSS	EXT_NSS
51	PA5	I/O	SPI1_SCK	DRV_SCK
52	PA6	I/O	ADC2_INP3	SOB
53	PA7	I/O	TIM3_CH2	A_UI_ENC
54	PC4	I/O	ADC1_INP4	SOA
55	PC5 *	I/O	GPIO_Output	DRV_CAL
56	PB0	I/O	LTDC_R3	
57	PB1	I/O	GPIO_EXTI1	I_ENC
58	PB2	I/O	QUADSPI_CLK	
59	PF11	I/O	FMC_SDNRAS	
60	PF12	I/O	FMC_A6	
61	VSS	Power		
62	VDD	Power		
63	PF13	I/O	FMC_A7	
64	PF14	I/O	FMC_A8	
65	PF15	I/O	FMC_A9	
66	PG0	I/O	FMC_A10	
67	PG1	I/O	FMC_A11	
68	PE7	I/O	FMC_D4	
69	PE8	I/O	FMC_D5	
70	PE9	I/O	FMC_D6	
71	VSS	Power		
72	VDD	Power		
73	PE10	I/O	FMC_D7	
74	PE11	I/O	FMC_D8	

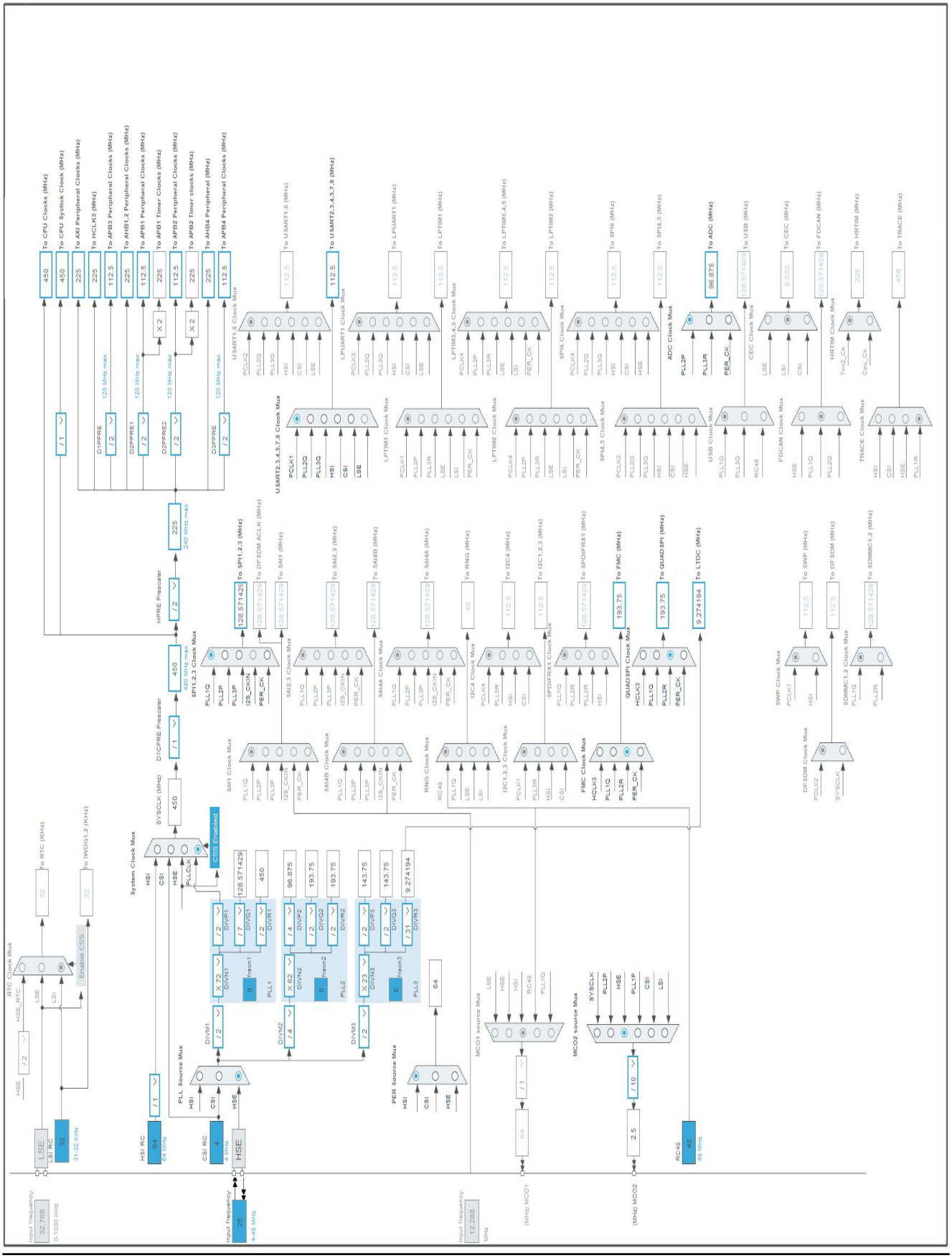
Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
75	PE12	I/O	FMC_D9	
76	PE13	I/O	FMC_D10	
77	PE14	I/O	FMC_D11	
78	PE15	I/O	FMC_D12	
79	PB10	I/O	USART3_TX	
80	PB11	I/O	USART3_RX	
81	VCAP	Power		
82	VDD	Power		
83	PH6	I/O	GPIO_EXTI6	FAN_FAULT
84	PH7 *	I/O	GPIO_Output	DIR
85	PH8	I/O	LTDC_R2	
86	PH9	I/O	TIM12_CH2	STEP
87	PH10	I/O	TIM5_CH1	KLV_LED
88	PH11	I/O	LTDC_R5	
89	PH12	I/O	LTDC_R6	
90	VSS	Power		
91	VDD	Power		
92	PB12	I/O	TIM1_BKIN	DRV_NFAULT
93	PB13	I/O	TIM1_CH1N	INLA
94	PB14	I/O	TIM1_CH2N	INLB
95	PB15	I/O	TIM1_CH3N	INLC
96	PD8	I/O	FMC_D13	
97	PD9	I/O	FMC_D14	
98	PD10	I/O	FMC_D15	
99	PD11 *	I/O	GPIO_Output	DC_EN
100	PD12	I/O	TIM4_CH1	B_PED_ENC
101	PD13	I/O	QUADSPI_BK1_IO3	
102	VSS	Power		
103	VDD	Power		
104	PD14	I/O	FMC_D0	
105	PD15	I/O	FMC_D1	
106	PG2 *	I/O	GPIO_Input	DC_IN
107	PG3 *	I/O	GPIO_Output	DRV_ENN
108	PG4	I/O	FMC_BA0	
109	PG5	I/O	FMC_BA1	
110	PG6	I/O	LTDC_R7	
111	PG7	I/O	LTDC_CLK	
112	PG8	I/O	FMC_SDCLK	
113	VSS	Power		

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
114	VDD33_USB	Power		
115	PC6	I/O	TIM3_CH1	B_UI_ENC
116	PC7 *	I/O	GPIO_Input	DIAG0
117	PC8 *	I/O	GPIO_Input	DIAG1
118	PC9	I/O	RCC_MCO_2	
119	PA8	I/O	TIM1_CH1	INHA
120	PA9	I/O	TIM1_CH2	INHB
121	PA10	I/O	TIM1_CH3	INHC
122	PA11	I/O	LTDC_R4	
123	PA12	I/O	SPI2_SCK	ENC_SCK
124	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
125	VCAP	Power		
126	VSS	Power		
127	VDD	Power		
128	PH13	I/O	LTDC_G2	
129	PH14	I/O	LTDC_G3	
130	PH15	I/O	LTDC_G4	
131	PI0 *	I/O	GPIO_Output	ENC_NSS
132	PI1 *	I/O	GPIO_Output	DCO
133	PI2	I/O	LTDC_G7	
134	PI3 *	I/O	GPIO_Output	SPI_MODE
135	VSS	Power		
136	VDD	Power		
137	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
138	PA15 (JTDI)	I/O	DEBUG_JTDI	
139	PC10	I/O	SPI3_SCK	EXT_SCK
140	PC11	I/O	SPI3_MISO	EXT_MISO
141	PC12	I/O	SPI3_MOSI	EXT_MOSI
142	PD0	I/O	FMC_D2	
143	PD1	I/O	FMC_D3	
144	PD2	I/O	GPIO_EXTI2	FRW_BUT
145	PD3	I/O	GPIO_EXTI3	REV_BUT
146	PD4 *	I/O	GPIO_Output	LED2
147	PD5 *	I/O	GPIO_Output	LED1
148	VSS	Power		
149	VDD	Power		
150	PD6	I/O	LTDC_B2	
151	PD7	I/O	SPI1_MOSI	DRV_MOSI
152	PG9	I/O	SPI1_MISO	DRV_MISO

Pin Number LQFP176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
153	PG10 *	I/O	GPIO_Output	DRV_NSS
154	PG11	I/O	LTDC_B3	
155	PG12	I/O	LTDC_B1	
156	PG13 *	I/O	GPIO_Output	QSPI_RST
158	VSS	Power		
159	VDD	Power		
160	PG15	I/O	FMC_SDNCAS	
161	PB3 (JTDO/TRACESWO)	I/O	DEBUG_JTDO-SWO	
162	PB4 (NJTRST)	I/O	DEBUG_JTRST	
163	PB5 *	I/O	GPIO_Output	LTDC_STNDB
164	PB6	I/O	QUADSPI_BK1_NCS	
165	PB7	I/O	TIM4_CH2	A_PED_ENC
166	BOOT0	Boot		
167	PB8	I/O	LTDC_B6	
168	PB9	I/O	LTDC_B7	
169	PE0	I/O	FMC_NBL0	
170	PE1	I/O	FMC_NBL1	
171	PDR_ON	Power		
172	VDD	Power		
173	PI4	I/O	LTDC_B4	
174	PI5	I/O	TIM8_CH1	BUZZER
175	PI6 *	I/O	GPIO_Output	VIB
176	PI7 *	I/O	GPIO_Input	TEST

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H743/753
MCU	STM32H743IITx
Datasheet	DS12110_Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

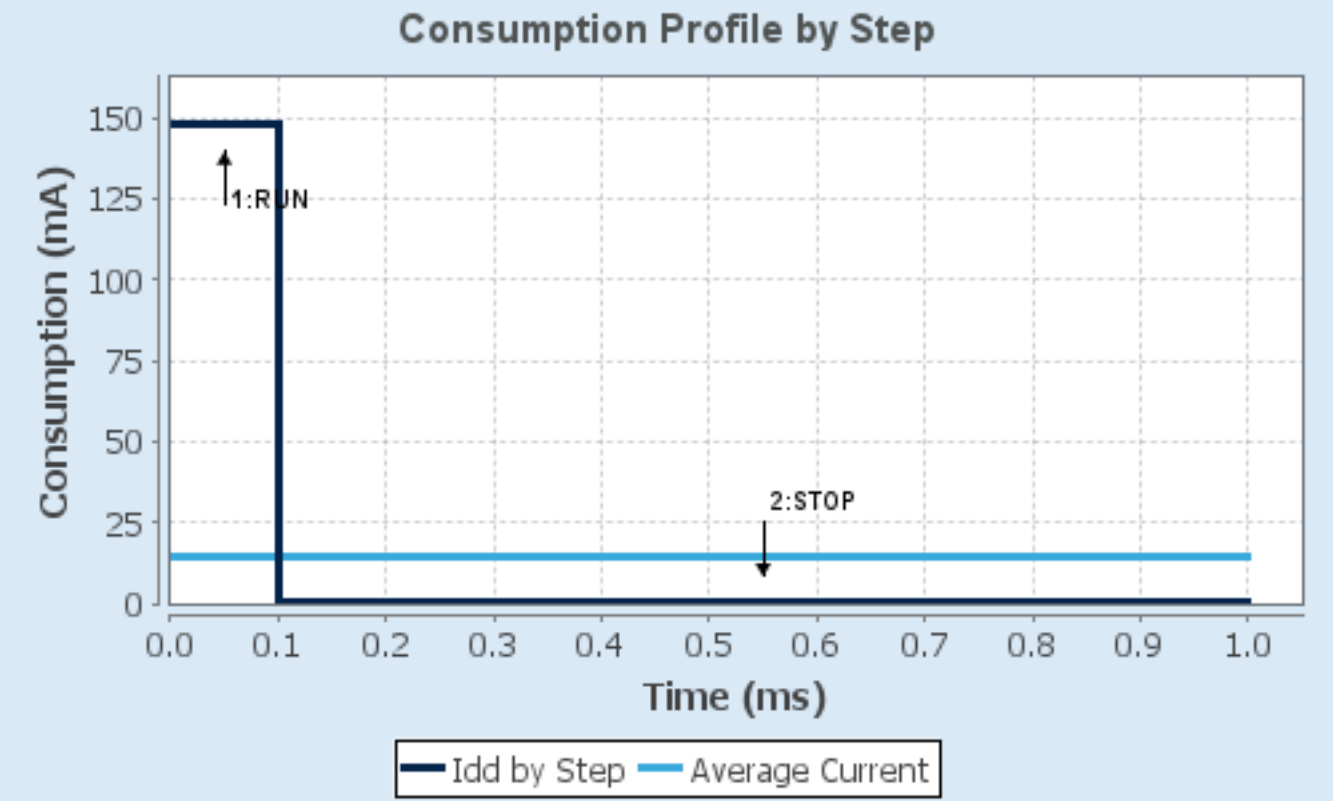
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 µA
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.91	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	FRW4_0_RTOS
Project Folder	C:\GitHub\Open-loop\FRW4_0_RTOS
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	Yes

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC1_Init	ADC1
5	MX_ADC2_Init	ADC2
6	MX_ADC3_Init	ADC3
7	MX_QUADSPI_Init	QUADSPI
8	MX_FMC_Init	FMC
9	MX_LTDC_Init	LTDC
10	MX_DMA2D_Init	DMA2D
11	MX_SPI1_Init	SPI1

Rank	Function Name	Peripheral Instance Name
12	MX_SPI2_Init	SPI2
13	MX_SPI3_Init	SPI3
14	MX_TIM1_Init	TIM1
15	MX_TIM2_Init	TIM2
16	MX_TIM3_Init	TIM3
17	MX_TIM4_Init	TIM4
18	MX_TIM8_Init	TIM8
19	MX_TIM5_Init	TIM5
20	MX_USART3_UART_Init	USART3
21	MX_CRC_Init	CRC
22	MX_TIM12_Init	TIM12
24	MX_TouchGFX_Init	STMicroelectronics.X-CUBE-TOUCHGFX.4.26.0
25	MX_TouchGFX_Process	STMicroelectronics.X-CUBE-TOUCHGFX.4.26.0

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN4: IN4 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	**Asynchronous clock mode divided by 2 ***
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Oversampling Ratio	1
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Rank	1
Channel	Channel 4
Sampling Time	**387.5 Cycles ***
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions	**Enable ***
Enable Injected Oversampling	Disable
Number Of Conversions	**1 ***
External Trigger Source	**Timer 1 Trigger Out 2 event ***
External Trigger Conversion Edge	Trigger detection on the rising edge
Injected Conversion Mode	None
Injected Queue	Injected Queue Disable
Rank	1
Channel	Channel 4

Sampling Time	**64.5 Cycles ***
Offset Number	No offset
Offset Signed Saturation	Disable
Injected Offset Saturation	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.2. ADC2

IN3: IN3 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	**Asynchronous clock mode divided by 2 ***
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	**Disable ***
Oversampling Ratio	1

ADC_Injected_ConversionMode:

Enable Injected Conversions	**Enable ***
Enable Injected Oversampling	Disable
Number Of Conversions	**1 ***
External Trigger Source	**Timer 1 Trigger Out 2 event ***
External Trigger Conversion Edge	Trigger detection on the rising edge
Injected Conversion Mode	None

Injected Queue	Injected Queue Disable
Rank	1
Channel	Channel 3
Sampling Time	**64.5 Cycles ***
Offset Number	No offset
Offset Signed Saturation	Disable
Injected Offset Saturation	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. ADC3

IN3: IN3 Single-ended

3.3.1. Parameter Settings:

ADC_Settings:

Resolution	ADC 16-bit resolution
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	**Disable ***
Oversampling Ratio	1

ADC_Injected_ConversionMode:

Enable Injected Conversions	**Enable ***
Enable Injected Oversampling	Disable
Number Of Conversions	**1 ***
External Trigger Source	**Timer 1 Trigger Out 2 event ***
External Trigger Conversion Edge	Trigger detection on the rising edge
Injected Conversion Mode	None
Injected Queue	Injected Queue Disable

Rank	1
Channel	Channel 3
Sampling Time	**64.5 Cycles ***
Offset Number	No offset
Offset Signed Saturation	Disable
Injected Offset Saturation	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.4. CORTEX_M7

3.4.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	**Enabled ***

Cortex Interface Settings:

CPU ICache	**Enabled ***
CPU DCache	**Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	**0x0 ***
MPU Region Size	4GB
MPU SubRegion Disable	**0x87 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x08000000 ***

MPU Region Size	**2MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	**level 1 ***
MPU Access Permission	**Privileged READS Permissions ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x90000000 ***
MPU Region Size	**256MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	**level 1 ***
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	**ENABLE ***
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0xC0000000 ***
MPU Region Size	**8MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	**ENABLE ***

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.5. CRC

mode: Activated

3.5.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.6. DEBUG

Debug: JTAG (5 pins)

3.7. DMA2D

mode: Activated

3.7.1. Parameter Settings:

Basic Parameters:

Transfer Mode	**Register to Memory ***
Color Mode	ARGB8888
Output Offset	0

3.8. FMC

SDRAM 1

Clock and chip enable: SDCKE0+SDNE0

Internal bank number: 4 banks

Address: 12 bits

Data: 16 bits

Byte enable: 16-bit byte enable

3.8.1. SDRAM 1:

SDRAM control:

Bank	SDRAM bank 1
Number of column address bits	8 bits
Number of row address bits	12 bits
CAS latency	**3 memory clock cycles ***
Write protection	Disabled
SDRAM common clock	**2 HCLK clock cycles ***
SDRAM common burst read	Disabled
SDRAM common read pipe delay	**1 HCLK clock cycle ***

SDRAM timing in memory clock cycles:

Load mode register to active delay	**2 ***
Exit self-refresh delay	**7 ***
Self-refresh time	**4 ***
SDRAM common row cycle delay	**6 ***
Write recovery time	**2 ***
SDRAM common row precharge delay	**2 ***
Row to column delay	**2 ***

3.8.2. Bank Mapping:

Mapping parameters:

FMC bank mapping	Default mapping

3.9. LTDC

Display Type: RGB888 (24 bits)

3.9.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	**41 ***
Horizontal Back Porch	**2 ***
Active Width	**480 ***
Horizontal Front Porch	**2 ***
HSync Width	40
Accumulated Horizontal Back Porch Width	42
Accumulated Active Width	522
Total Width	524

Synchronization for Height:

Vertical Synchronization Height	**10 ***
Vertical Back Porch	2
Active Height	**272 ***
Vertical Front Porch	**8 ***
VSync Height	9
Accumulated Vertical Back Porch Height	11
Accumulated Active Height	283
Total Height	291

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	**Active High ***
Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.9.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha

	255 *
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0

Number of Layers:

Number of Layers	**1 layer ***

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	**480 ***
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	**272 ***

Pixel Parameters:

Layer 0 - Pixel Format	ARGB8888

Blending:

Layer 0 - Alpha constant for blending	**255 ***
Layer 0 - Blending Factor1	Alpha constant
Layer 0 - Blending Factor2	Alpha constant

Frame Buffer:

Layer 0 - Color Frame Buffer Start Adress	**0xC0000000 ***
Layer 0 - Color Frame Buffer Line Length (Image Width)	**480 ***
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	**272 ***

3.10. MEMORYMAP

mode: Activated

3.11. QUADSPI

QuadSPI Mode: Bank1 with Quad SPI Lines

3.11.1. Parameter Settings:

General Parameters:

Clock Prescaler	**3 ***
Fifo Threshold	**4 ***
Sample Shifting	No Sample Shifting
Flash Size	**25 ***
Chip Select High Time	**2 Cycles ***

Clock Mode	Low
Flash ID	Flash ID 1
Dual Flash	Disabled

3.12. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

mode: Master Clock Output 2

3.12.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	3 WS (4 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL2 input frequency range	Between 4 and 8 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	**MEDIUM VCO range ***

3.13. SPI1

Mode: Full-Duplex Master

3.13.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	**16 Bits ***

First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	**16 ***
Baud Rate	**8.035714 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	**2 Edge ***

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.14. SPI2

Mode: Full-Duplex Master

3.14.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	**16 Bits ***
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	**16 ***
Baud Rate	**8.035714 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	**2 Edge ***

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data

Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.15. SPI3

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.15.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	**16 ***
Baud Rate	**8.035714 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.16. SYS

Timebase Source: TIM6

3.17. TIM1

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Channel4: PWM Generation No Output

Activate-Break-Input: Activate Break Input

3.17.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**1 ***
Counter Mode	**Center Aligned mode3 ***
Counter Period (AutoReload Register - 16 bits value)	**2812 ***
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	**Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	**Output Compare (OC4REF) ***

Break And Dead Time management - BRK Configuration:

BRK State	**Disable ***
BRK Polarity	**Low ***
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Enable
Digital Input Polarity	Polarity High
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable

- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off
Dead Time	**30 ***

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	**1406-1 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
Pulse (16 bits value)	**1406-1 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	**1406-1 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	**2100 ***
Output compare preload	Enable

Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.18. TIM2

Combined Channels: Encoder Mode

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	**Encoder Mode TI1 and TI2 ***
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.19. TIM3

Combined Channels: Encoder Mode

3.19.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	

	9999 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	**Falling Edge ***
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	**10 ***
____ Parameters for Channel 2 ____	
Polarity	**Falling Edge ***
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	**10 ***

3.20. TIM4

Combined Channels: Encoder Mode

3.20.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

____ Parameters for Channel 2 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.21. TIM5

Channel1: PWM Generation CH1

3.21.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**124 ***
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	**99 ***
Internal Clock Division (CKD)	No Division
auto-reload preload	**Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (32 bits value)	**49 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.22. TIM8

Channel1: PWM Generation CH1

3.22.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division

Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.23. TIM12

Channel2: PWM Generation CH2

3.23.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.24. USART3

Mode: Asynchronous

3.24.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.25. FREERTOS

Interface: CMSIS_V2

3.25.1. Config parameters:

API:

FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled

USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**262144 ***
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	**Enabled ***
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	**Option2 ***

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.25.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.25.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	**Enabled ***

Project settings (see parameter description first):

Use FW pack heap file	Enabled

3.26. STMicroelectronics.X-CUBE-TOUCHGFX.4.26.0 mode: GraphicsJjApplication

3.26.1. TouchGFX Generator:

Display:

Interface	**Parallel RGB (LTDC) ***
Framebuffer Pixel Format (LTDC)	ARGB8888
Width (LTDC)	480

Height (LTDC)	272
Use Larger Framebuffer Stride	**Yes ***
Framebuffer Stride Value (LTDC)	480
Framebuffer Strategy	**Double Buffer ***
Buffer Location	**By Address ***
Start Address	**0xC0000000 ***
Start Address 2	**0xC007F800 ***

Driver:

Application Tick Source	Custom
Use DMA2D Accelerator (ChromART)	**Yes ***
Real-Time Operating System	CMSIS_RTOS_V2

Additional Features:

Vector Rendering	Disabled

Video Decoding:

Type	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC4	ADC1_INP4	Analog mode	No pull-up and no pull-down	n/a	SOA
ADC2	PA6	ADC2_INP3	Analog mode	No pull-up and no pull-down	n/a	SOB
ADC3	PF7	ADC3_INP3	Analog mode	No pull-up and no pull-down	n/a	SOC
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
	PA15 (JTDI)	DEBUG_JTDI	n/a	n/a	n/a	
	PB3 (JTDO/TRACESWO)	DEBUG_JTDO-SWO	n/a	n/a	n/a	
	PB4 (NJTRST)	DEBUG_JTRST	n/a	n/a	n/a	
FMC	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC0	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC3_C	FMC_SDCKE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH3	FMC_SDNE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE0	FMC_NBL0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE1	FMC_NBL1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
LTDC	PE4	LTDC_B0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PE5	LTDC_G0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PE6	LTDC_G1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PI9	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PI10	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PI11	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PF10	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA2	LTDC_R1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PH2	LTDC_R0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PH4	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA3	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB0	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High**	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PH8	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PH11	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PH12	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PG6	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PG7	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PA11	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PH13	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PH14	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PH15	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PI2	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PD6	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PG11	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PG12	LTDC_B1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PB8	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PB9	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PI4	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
QUADSPI	PE2	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	
	PF8	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High** *	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PF9	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB2	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PD13	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB6	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
	PC9	RCC_MCO_2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	DRV_SCK
	PD7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	DRV_MOSI
	PG9	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	DRV_MISO
SPI2	PC1	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC_MOSI
	PC2_C	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC_MISO
	PA12	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC_SCK
SPI3	PA4	SPI3_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_NSS
	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_SCK
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_MISO
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	EXT_MOSI
TIM1	PB12	TIM1_BKIN	Alternate Function Open Drain	No pull-up and no pull-down	Low	DRV_NFAULT
	PB13	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Low	INLA
	PB14	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	INLB
	PB15	TIM1_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Low	INLC
	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	INHA
	PA9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	INHB
	PA10	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	INHC
TIM2	PA0	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	A_ENC
	PA1	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	B_ENC
TIM3	PA7	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	A_UI_ENC
	PC6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	B_UI_ENC
TIM4	PD12	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	B_PED_ENC
	PB7	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	A_PED_ENC
TIM5	PH10	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	KLV_LED
TIM8	PI5	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	BUZZER

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
TIM12	PH9	TIM12_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	STEP
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED0
	PI8	GPIO_EXTI8	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	SIDE_BUTTON
	PC13	GPIO_EXTI13	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	BACK_BUTTON
	PC14-OSC32_IN (OSC32_IN)	GPIO_EXTI14	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	LIMIT_BUTTON
	PC15-OSC32_OUT	GPIO_EXTI15	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	UI_ENC_BUTTON
	PF6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DRV_EN
	PH5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	VREF_EN
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DRV_CAL
	PB1	GPIO_EXTI1	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	I_ENC
	PH6	GPIO_EXTI6	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	FAN_FAULT
	PH7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIR
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DC_EN
	PG2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DC_IN
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DRV_ENN
	PC7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DIAG0
	PC8	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DIAG1
	PI0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ENC_NSS
	PI1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DCO
	PI3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI_MODE
	PD2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	FRW_BUT
	PD3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	REV_BUT
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED2
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED1
	PG10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DRV_NSS

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PG13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	QSPI_RST
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LTDC_STNDB
	PI6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	VIB
	PI7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	TEST

4.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI3_RX	DMA1_Stream2	Peripheral To Memory	Low
SPI3_TX	DMA1_Stream3	Memory To Peripheral	Low
SPI2_RX	DMA1_Stream4	Peripheral To Memory	Low
SPI2_TX	DMA1_Stream5	Memory To Peripheral	Low

SPI3_RX: DMA1_Stream2 DMA request Settings:

Mode:	**Circular ***
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	**Half Word ***
Memory Data Width:	**Half Word ***

SPI3_TX: DMA1_Stream3 DMA request Settings:

Mode:	**Circular ***
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	**Half Word ***
Memory Data Width:	**Half Word ***

SPI2_RX: DMA1_Stream4 DMA request Settings:

Mode:	**Circular ***
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	**Half Word ***
Memory Data Width:	**Half Word ***

SPI2_TX: DMA1_Stream5 DMA request Settings:

Mode:	**Circular ***
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	**Half Word ***
Memory Data Width:	**Half Word ***

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 stream2 global interrupt	true	5	0
DMA1 stream3 global interrupt	true	5	0
DMA1 stream4 global interrupt	true	5	0
DMA1 stream5 global interrupt	true	5	0
ADC1 and ADC2 global interrupts	true	5	0
EXTI line[9:5] interrupts	true	5	0
TIM3 global interrupt	true	5	0
SPI2 global interrupt	true	5	0
EXTI line[15:10] interrupts	true	5	0
SPI3 global interrupt	true	5	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	15	0
LTDC global interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line3 interrupt	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM4 global interrupt	unused		
SPI1 global interrupt	unused		
USART3 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 update interrupt and TIM13 global	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
interrupt			
TIM8 trigger and commutation interrupts and TIM14 global interrupt	unused		
TIM8 capture compare interrupt	unused		
FMC global interrupt	unused		
TIM5 global interrupt	unused		
FPU global interrupt	unused		
LTDC global error interrupt	unused		
DMA2D global interrupt	unused		
QUADSPI global interrupt	unused		
HSEM1 global interrupt	unused		
ADC3 global interrupt	unused		

4.5.2. NVIC Code generation

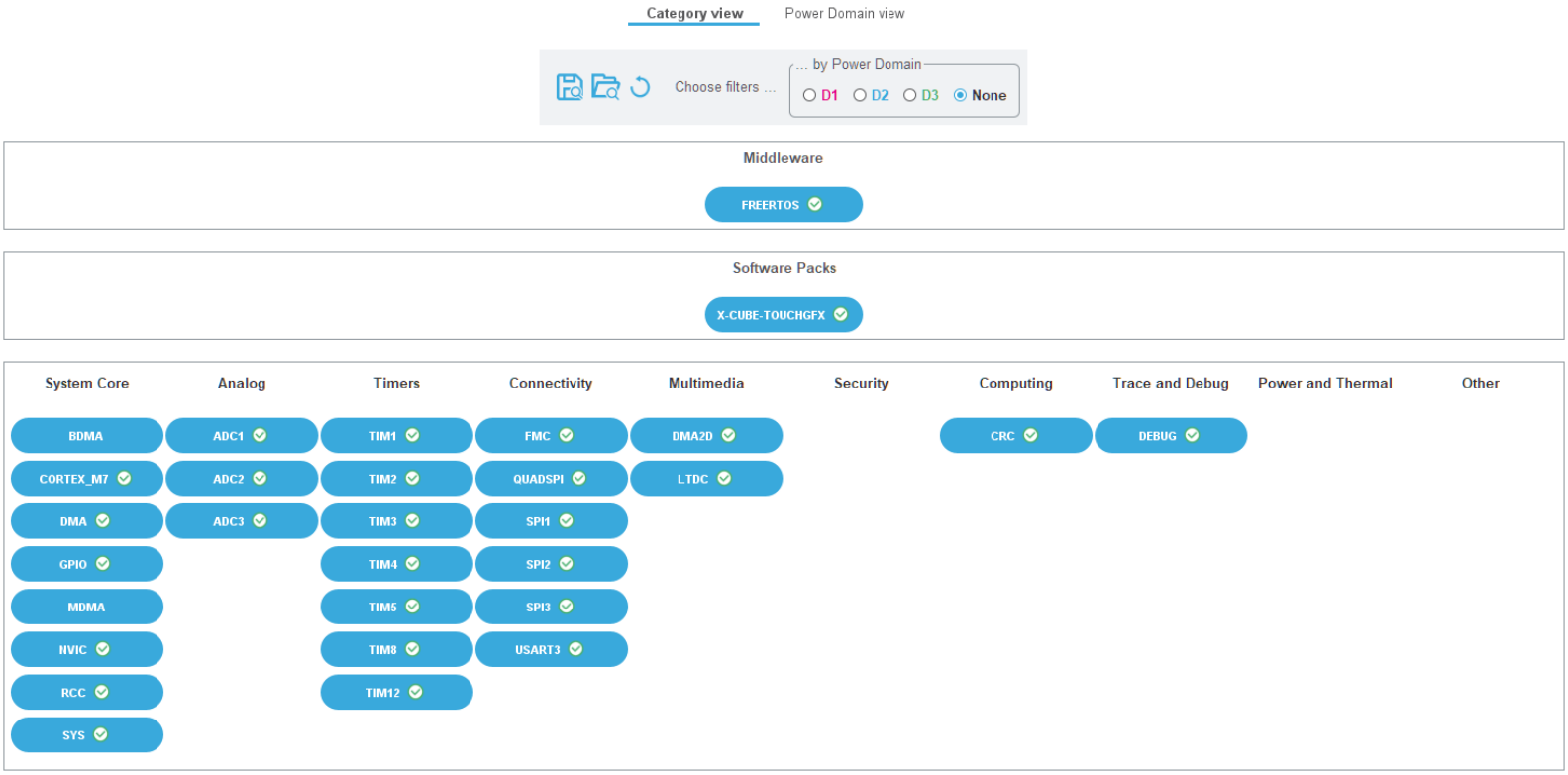
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	true
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 stream2 global interrupt	false	true	true
DMA1 stream3 global interrupt	false	true	true
DMA1 stream4 global interrupt	false	true	true
DMA1 stream5 global interrupt	false	true	true
ADC1 and ADC2 global interrupts	false	true	true
EXTI line[9:5] interrupts	false	true	true
TIM3 global interrupt	false	true	true
SPI2 global interrupt	false	true	true
EXTI line[15:10] interrupts	false	true	true
SPI3 global interrupt	false	true	true
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	false	true	true
LTDC global interrupt	false	true	true

*** User modified value**

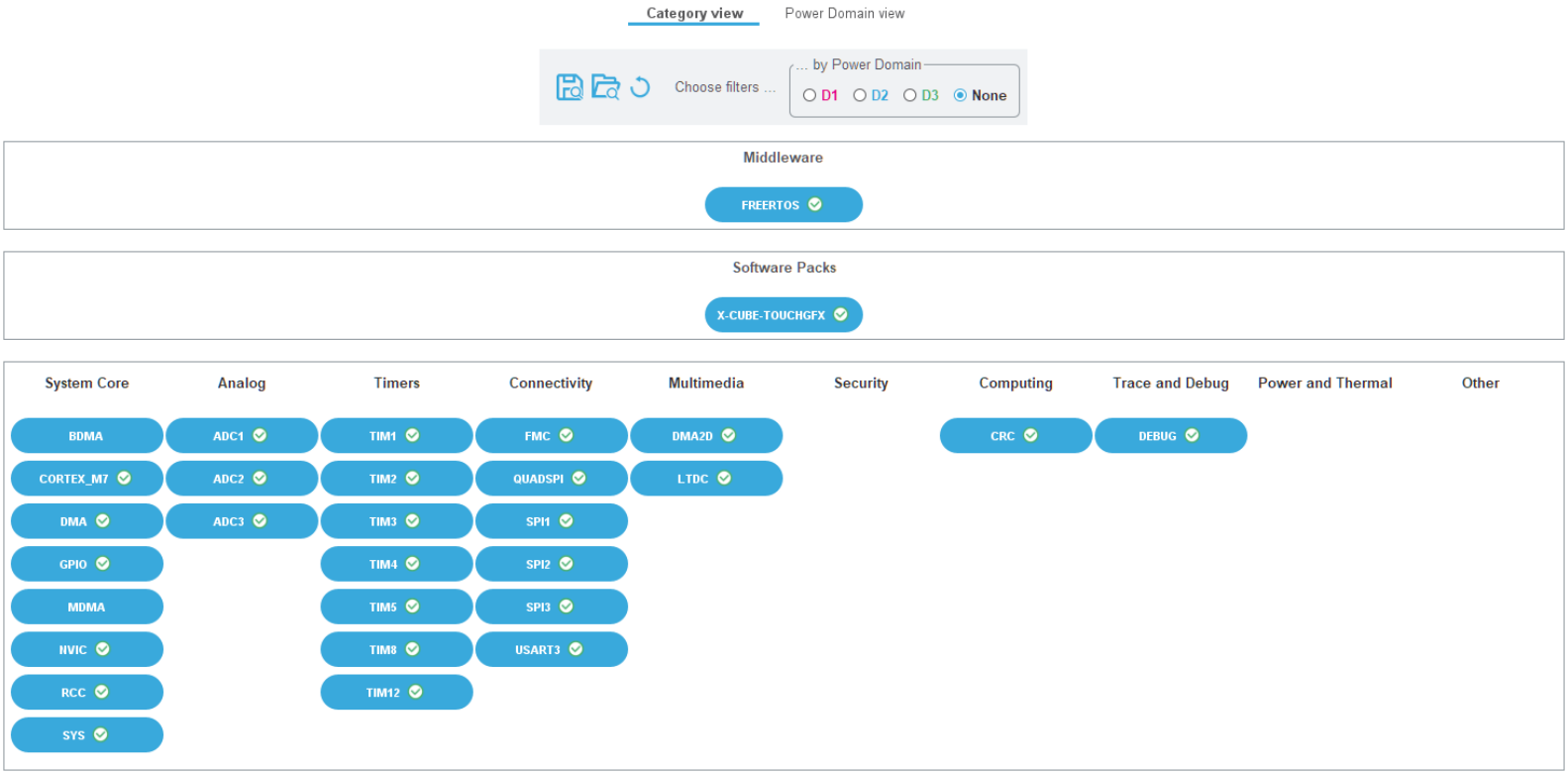
5. System Views

5.1. Category view

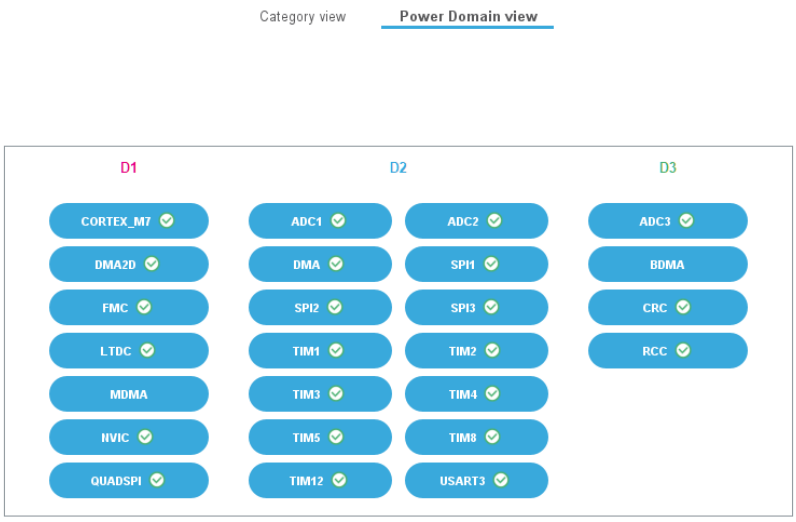
5.1.1. Current



5.1.2. Without filters



5.2. Power Domain view



6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-TOUCHGFX	4.26.0	Class : Graphics Group : Application Variant : TouchGFX Generator Version : 4.26.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4936-migration-of-microcontroller-applications-from-stm32f7-series-to-stm32h743753-line-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4938-getting-started-with-stm32h74xig-and-stm32h75xig-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4943-how-to-use-chromart-accelerator-to-refresh-an-lcdtft-display-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5507-how-to-use-crc-to-check-the-integrity-of-the-internal-flash-memory-on-stm32h7-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5337-guidelines-for-estimating-stm32h7-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4996-how-to-use-the-jpeg-codec-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-

	recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5200-getting-started-with-the-stm32h7-mcu-sdmmc-host-controller-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
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